INCH-POUND

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29 August 2019
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AMENDMENT 3
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MILITARY SPECIFICATION

ADHESIVE SYSTEM, EPOXY-ELASTOMERIC, FOR GLASS TO METAL

Reactivated after 29 August 2019 and may be used for new and existing designs and acquisitions.

1. SCOPE

1.1 Scope. This amendment covers elastomer modified epoxy resin adhesive bonding systems for the structural joining of optical glass prisms to metal, the sealing of glass and metal components, and for other applications requiring bonding of a similar nature. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

1.2 Classification. Adhesives bonding systems are of the following types, as specified (see 6.2)

Type I - System composed of an epoxy primer component and an epoxy adhesive component.

Type II - System composed of an epoxy adhesive component.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

AMSC N/A FSC 8040

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/SAE AMS 4035 - Aluminum Alloy Plate & Sheet (2024)

(Copies may be ordered online at https://www.ansi.org)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1640 - Standard Test Method for Drying, Curing, or Film Formation of Organic Coatings at Room Temperature

(Copies may be ordered online at https://www.astm.org)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtain (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Approval. The adhesive furnished under this specification shall be a product which has been tested, and passed the approval tests specified herein.

3.2 Materials.

3.2.1 Components. The components of the Type I adhesive system shall consist of a primer and an epoxy-elastomer adhesive. The components of the Type II adhesive system shall consist of the epoxy-elastomer adhesive only.

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3.2.2 Adhesive. The adhesive shall be a thermosetting resin and shall be based on epichlorohydrin-bisphenol A-Type epoxy resin, modified with acrylonitrile-butadiene rubber.

3.2.3 Primer. The Primer shall be a two component (Parts A and B) room temperature curing liquid epoxy and shall meet the physical properties of this specification.

3.3 Bond Strength (Type I and Type II). The bond strength for the Type I and Type II adhesive bonding systems shall be as specified in Table I.

TABLE I. Bond Strength.

TEST NO.	PROPERTY	TEST CONDITIONS	NO. OF SPECIMENS	TEST TEMPERTURE	PARA. REF.	MIN AVG. STRENGTH REQT.	
						TYPE I	TYPE II
1.	Tensile Shear	23° ± 2°C, 50 ± 10% R.H.	6	23° ± 2°C	4.5.4.1	11 MPa	11 MPa
2.	Tensile Shear	60 minutes at 71° ± 2°C	6	71° ± 2°C	4.5.4.2	5.86 MPa	
3.	Tensile Shear	60 minutes at -51° ± 2°C	6	-51° ± 2°C	4.5.4.3	13.8 MPa	
4.	Tensile	10 days at at 71° ± 2°C & 95-100% R.H.	6	23° ± 2° C	4.5.4.4	11 MPa	

3.4 Low Temperature thermal stability (Type I and Type II). Bonded prisms shall not become detached or cracked when tested at -62° ± 2°C for 4 hours.

3.5 Mechanical shock stability (Type I only).

3.5.1 Mechanical shock stability- low temperature. Bonded prism assemblies shall not become detached during shock testing at -43° ± 2°C. The bonded assemblies shall be shocked at 250, 300, 350, and 400 g's. The shock pulse shall have a wave form approximating a half-sine with a minimum time duration of 1.5 milliseconds.

3.5.2 Mechanical shock stability - high Humidity. Bonded prism assemblies shall not become detached during shock testing after exposure to 3 temperature humidity cycles. Each cycle shall consist of exposure to 95-100% relative humidity at 70° ± 3°C for 24 hours. The bonded assembles shall be shocked at 250, 300, 350, and 400 g's. The shock pulse shall have a wave form approximating a half-sine with a minimum time duration of 1.5 milliseconds.

3.6 Storage life. The Type I and Type II adhesive bonding systems shall meet the bond strength requirements of tests No. 1, 2, and 3 of Table I, if applicable, after storage for 12 months at a temperature not to exceed 30° ± 2°C in the absence of sunlight.

3.7 Product characteristics.

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3.7.1 Adhesive component.

3.7.1.1 Application life. The application life of the adhesive when mixed and ready for use at 23° ± 3°C, shall be a minimum of 30 minutes.

3.7.1.2 Curing time and temperature. When subjected to a temperature not to exceed 74°C at the bond line, the adhesive shall cure in 4 hours maximum. At a temperature of 25° ± 2°C, the cure time shall not exceed 7 days.

3.7.1.3 Curing pressure. The adhesive shall be capable of curing and making satisfactory bonded joints as described within this specification when only contact pressure is applied during the period of cure.

3.7.2 Primer component.

3.7.2.1 Viscosity. The viscosity of the admixed primer (i.e. primer plus curing solution) one hour after mixing shall be 15 to 20 seconds using a No. 2 Zahn cup.

3.7.2.2 Drying time. The admixed primer shall set to touch in not more than 60 minutes and shall be dried hard in not more than 24 hours.

3.7.2.3 Mixing. Equal volumes of Component A and Component B shall show easy mixing to a smooth homogeneous product.

3.7.2.4 Pot life. The admixed primer shall have a minimum useable pot life of 8 hours at 25°C.

3.8 Instruction sheet. The manufacturer shall provide a dated, coded and titled instruction sheet with each shipment of adhesive outlining instruction for its use. The instruction sheet shall include the following information:

a. Complete recommended pre-bonding treatments and cleaning processes.

b. Maximum useable storage and pot life of the adhesive and primer. The pot life of the adhesive after the addition of the activator shall be based on definite quantities of the adhesive.

c. Mixing instructions including recommended percentage of activator and any temperature control necessary when mixing the adhesive or primer.

d. Application instructions, including application temperature and relative humidity, acceptable thickness range, and method of cleaning equipment.

e. Typical cure conditions for the adhesive and primer.

f. Necessary related data and precautions to be observed throughout all operations,

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including:

1. Standard operating procedure (SOP) in the use of the material.

2. Physical data on the characteristics of the components (boiling point and flash point).

3. Fire and explosion hazard data including guidance on the proper protection to use if a fire occurs in the area.

4. Reactivity data - guidance on classes of material which are not compatible with the components in the epoxy.

5. Precautions to take in the event of a spill or leak.

g. Any other pertinent information relative to the use and storage of the adhesive and or primer.

4. VERIFICATION

4.1 General provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection and testing of the adhesive shall be classified as follows:

a. Approval tests (see 4.5)

b. Quality conformance tests (see 4.6)

4.3 Test conditions.

4.3.1 Specimen conditions. All specimens shall be conditioned at 23° ± 2°C and 50 percent relative humidity for at least 24 hours. Other conditioning shall be prescribed in Table 1. Preparation of the test specimens shall be in accordance with 4.4 and as described herein.

4.3.2 Normal conditions (room temperature). Strength properties shall be determined for room temperature adhesive at 23° ± 2°C, no sooner than 10 minutes after specimens have reached equilibrium at such temperature.

4.3.3 Elevated temperature conditions. Conditioning of specimens for elevated temperature shear strength tests shall require a suitable oven to maintain the specimen at

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71° ± 2°C for a period of 60 minutes prior to testing.

4.3.4 Low temperature conditions. Conditioning of specimens for low temperature shear strength tests shall require a suitable conditioning chamber to maintain the specimens at -51° ± 2°C for a period of 60 minutes prior to testing.

4.3.5 Humidity conditions. Conditioning of specimens for humidity shear strength tests shall require a suitable conditioning chamber to maintain the specimen at 71° ± 2°C and 95 to 100% relative humidity for a period of 10 days prior to testing.

4.4 Preparation of test specimens.

4.4.1 Preparation of shear test specimens. Test specimens as shown in Figure I shall be prepared in sufficient quantities for all the necessary tests. Specimens for testing all the Table 1 properties shall be prepared from 0.32 cm (min) by 2.54 cm by 10 cm 2024-T3 aluminum conforming to ANSI/SAE AMS 4035; and glass microslides 2.54 cm by 2.54 cm by 0.30 cm (min). The cleaning, priming, and bonding procedures shall be in accordance the manufacturer's instruction sheet. The test specimens are to be assembled as shown in Figure I with the glass forming a bond of 6.45 sq. cm to one aluminum adherent and 3.23 sq. cm to the other aluminum adherent.

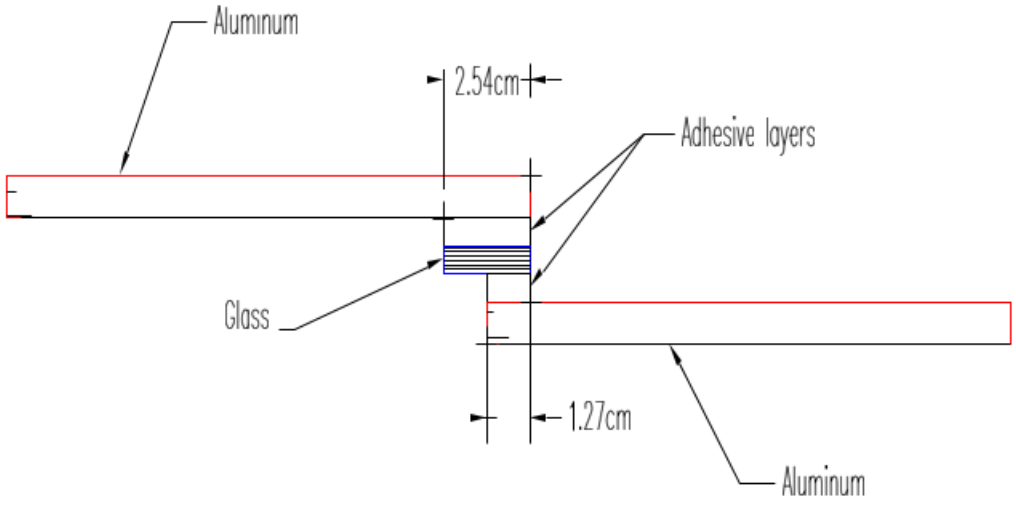


FIGURE 1. Assembly of shear specimens.

4.4.2 Preparation of thermal and mechanical shock stability test specimens. Test specimens shall be prepared in sufficient quantities for all the necessary tests. Specimens for testing the thermal and mechanical properties shall consist of right angle optical glass prisms 4 cm x 4 cm x 4 cm bonded to an aluminum plate 6 cm x 6 cm x 1.0 cm. The priming and bonding procedures shall be in accordance with the manufacturer's instruction sheet.

4.5 Approval tests.

4.5.1 Approval retest. Unless otherwise specified by the approving activity,

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adhesives which have previously passed the approval tests but havesubsequently been modified in any manner shall satisfactorily pass re-testing of the material in order to retain approval. Approval tests may be repeated on material previously found satisfactory at any time at the option of the activity responsible for approval.

4.5.2 Manufacturer's data. Approval samples shall be accompanied by a test report showing results of all tests required by this specification except the tests for bond strength, low temperature thermal stability and mechanical shock stability.

4.5.3 Instruction sheet. Two copies of the manufacturer's instructions for use of the adhesive shall be furnished with the approval samples (see 3.8).

4.5.4 Tests. The approval tests shall consist of all the tests of this specification.

4.5.4.1 Normal temperature tensile shear test. (Test No. 1 of Table I) The test specimen shall be gripped tightly and uniformly across the ends of the jaws of the test machine with the jaws and specimen so aligned that the jaws are directly opposite each other, and in such a position that an imaginary straight line will pass through the center of the bonded area and through the points of suspension. All specimens shall be pulled at a jaw separation rate of 0.13 cm per minute. All failing loads shall be expressed in MPa of actual shear area, calculated to the nearest 0.65 sq. cm. A total of six specimens shall be tested. The average strengths of these six specimens shall equal or exceed that specified for test number 1 in Table I. An average shear strength, less than that specified in Table I, shall cause rejection of the adhesive.

4.5.4.2 Elevated temperature tensile shear test. (Test No. 2 of Table I) The 71° ± 2°C shear strength test shall be in accordance with 4.5.4.1 with the added requirement that a suitable oven shall be provided to maintain the specimens at the specified test temperature (See 4.3.3 and Table I). Six specimens shall be tested for tensile shear test listed in Table I. The average shear strength of each set of specimens tested shall equal or exceed that specified for Test No. 2 in Table I. An average shear strength less than that specified in Table I shall cause rejection of the adhesive.

4.5.4.3 Low temperature tensile shear test. (Test No. 3 of Table I) The low temperature tensile shear test shall be in accordance with 4.5.4.1 with the added requirement that a suitable chamber shall be provided in maintaining the temperature at -51° ± 2°C (see 4.3.4). Six specimens shall be tested for tensile shear strength at -51° ± 2°C. The average shear strength of test specimens shall equal or exceed that specified for Test No. 3 of Table I. An average shear strength less than that specified in Table I shall cause rejection of the adhesive.

4.5.4.4 High humidity tensile shear test. (Test No. 4 of Table I) The humidity tensile shear test shall be in accordance with 4.5.4.1 after the specimens have been conditioned at 71° ± 2°C and 95 to 100% relative humidity. A total of six specimens shall be tested. The average strengths shall equal or exceed that specified for Test No. 4 in Table I. An average shear strength less than that specified in Table I shall cause rejection of the adhesive.

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4.5.4.5 Low temperature thermal stability. The adhesive shall be evaluated for its low temperature thermal stability by placing prism assemblies, prepared as specified in 4.4.2, in a cold box at room temperature and allowing the temperature to drop to -62.5° ± 2°C at a rate not to exceed 20°C per hour. The temperature shall be maintained for four hours, after which the specimens shall be returned to room temperature and examined to meet the requirements of 3.4. If one or more of the test specimens fails the test, the adhesive shall be rejected A total of five specimens shall be tested.

4.5.4.6 Mechanical shock stability.

4.5.4.6.1 Mechanical shock stability-low temperature. The adhesive shall be evaluated for its low temperature mechanical shock stability by testing prism assemblies, prepared as specified in 4.4.2 in the vertical plane, in accordance with the pulse shape, peak value and time duration as specified in 3.5.1. The specimens shall be refrigerated by placing them in a cold box at room temperature and allowing the temperature to drop to -51° ± 2°C and maintaining that temperature for two hours. The specimens shall then be removed from the cold box and mounted on a fixture of the shock test machine and then shock tested. This must be done expeditiously so that the temperature of the assembly does not increase above -40°C. The mounted prism shall first be shock tested at 250 g's and the shock test increased at increments of 50 g's until the 400 g level has been reached. One shock test shall be made at each level. If one or more of the test specimens fails the test, the adhesive shall be rejected. A total of five specimens shall be tested.

4.5.4.6.2 Mechanical shock stability - High humidity. The adhesive shall be evaluated for its mechanical shock stability by testing prism assemblies in accordance with the pulse shape, peak value and time duration as specified in 3.5.2. The specimens shall be placed in a humidity chamber at 71° ± 3°C and 95 to 100% relative humidity for 24 hours. The specimens shall be removed from the humidity chamber and mounted on the fixture of a shock test machine and shock tested. The mounted prism shall first be shock tested at 250 g's and the shock test increased at increments of 50 g's until 400 g's has been reached. Three shock tests shall be made at each interval. This combination of humidity and shock shall constitute one test cycle. A total of three cycles shall be performed. If one or more of the test specimens fails the test, the adhesive shall be rejected. A total of five specimens shall be tested.

4.5.4.7 Storage life tests. A .47 liter package of each component for Type I or Type II shall be stored under temperature and time conditions specified in 3.6. At the end of the storage period, the components shall be used in the preparation of bonded lap joint specimens as required by test nos., 1, 2, and 3 of Table I. Six specimens shall be tested at each of the applicable test conditions. The strength of the specimens shall be equal or exceed that specified for the applicable tests in Table I.

4.5.4.8 Application life. Approximately 10 grams of the mixed adhesive shall be placed in a suitable container such as a plastic dish. A spatula with a 1 cm wide flat top shall be used in this test. Stir the adhesive frequently with the spatula and note the time at which the adhesive is considered to be definitely no longer spreadable on a glass or metal surface with spatula. The time to reach this consistency after mixing is the application life

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of the adhesive at 23° ± 3°C. The adhesive shall meet the requirements of 3.7.1.1 for application life.

4.5.4.9 Primer viscosity. The viscosity of the primer shall be determined by using a number 2 Zahn cup. The cup shall be immersed in the admixed primer one hour after mixing. The temperature of the primer shall be maintained at 25° ± 2°C. After allowing bubbles to disappear when the cup is immersed in the primer, the cup shall be removed from the liquid and simultaneously the stop watch shall be started. At the first distinct break in the stream, the stop watch shall be stopped. The average elapsed time of three such readings shall be in conformance with 3.7.2.1.

4.5.4.10 Primer drying time. The drying time shall be determined in accordance with ASTM D1640 during preparation of specimens specified in 4.4 to determine compliance with 3.7.2.2.

4.5.4.11 Primer mixing. Stir part A until completely uniform. Add an equal volume of part B, stir, and examine for conformity to the requirements of 3.7.2.3.

4.5.4.12 Pot life. The admixed primer shall be brushable after 8 hours at 25° ± 2°C to establish compliance with 3.7.2.4. The primer may be kept covered during this test to minimize evaporation of solvent.

4.6 Quality conformance tests. Quality conformance tests shall consist of the sampling tests (4.6.1) and the inspection of filled containers (4.6.2).

4.6.1 Sampling for inspection and acceptance.

4.6.1.1 Adhesive. One container of adhesive shall be selected at random from each lot. The adhesive shall be subjected to the tests in 4.5.4.8. If the sample fails any tests, the lot shall be rejected.

4.6.1.2 Primer. One container of primer shall be selected at random from each lot. The primer shall be subjected to the following determinations. If the sample fails any test, the lot shall be rejected.

a. Drying time (4.5.4.10)

b. Mixing (4.5.4.11)

4.6.1.3 Lot. A lot shall consist of all the material (adhesive or primer offered for delivery at one time).

4.6.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot in accordance with inspection Level III, Table II. The sample containers shall be subjected to the inspection of filled containers as specified in 4.7.

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4.7 Inspection of filled containers. Each sample filled containers selected in accordance with 4.6.2 shall be examined for defects of construction relative to the container and the closure, or evidence of leakage and for unsatisfactory markings. Each sample filled container shall be examined to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. (6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adhesive systems covered by this specification are intended primarily for use in bonding glass to metal. The materials to be bonded include glass prisms and other optical elements to their metal supports in optical fire control instruments. For maximum reliability and environmental resistance especially to humid conditions, the Type I adhesive system is recommended Both the Types I and II adhesive systems may be used for general purpose bonding of materials such as aluminum, stainless steel, brass, bronze, plastics, glass, etc., to themselves or in dissimilar combinations. Various bond line thickness may be used for bonded assemblies, however, for the structural bonding of glass prisms to metal mounts a bond line thickness of .036 to 0.41 cm is recommended. Minor assemblies bonded with the adhesive systems of this specification should be thoroughly evaluated for the effect of the adhesive on the optical properties of the assemblies.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type require (See 1.2).

c. Quantity desire

d. Certified test report for this adhesive should be available upon request by the procuring activity.

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e. Package marking should contain the following information:

1. Specification and type.

2. Manufacturer's designation.

3. Date of manufacture.

4. Recommended expiration date.

5. Storage requirements.

6. Precautionary markings.

7. Manufacturer's name and address.

8. Manufacturer's batch Identification.

9. Certified test report for this adhesive should be available upon request by the procuring activity.

f. A copy of the manufacturer's instructions should be enclosed with or attached to each container of adhesive as specified (see 3.8).

6.3 Debonding procedure. Satisfactory debonding of glass to metal bonds, formed with epoxy based adhesives conforming to the requirements as contained herein, can be accomplished through the use of the solvent N-Methyl-2-PyrrolIone. A quantity of solvent, sufficient to completely submerse the assembly to be debonded, should be brought to an elevated temperature in excess of 90°C but less than 150°C. (The higher the solvent temperature the faster will be the debonding effect). The volatility of the solvent at elevated temperatures will necessitate a hooded area plus the use of condensing retort for the confining and condensing of gaseous solvent. A debonding effect should be obtained in one hour or less for a $2cm^2$ assembly and in three hours or less for a $9cm^2$ assembly depending on actual solvent temperature. The time of submersion may have to be extended to completely remove the primer coating (if applicable).

6.4 Previous source. Both types of this adhesive have successfully been sourced through the following supplier in the past:

EMS ACQUISITION CORP.
ELECTRON MICROSCOPY SCIENCES
1560 INDUSTRY RD
HATFIELD, PA 194403249

Phone: 215-412-8400
http://www.emsdiasum.com

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Test Reference: A48611A77

This is no guarantee that this supplier can still provide approved product, nor that the product couldn't be sourced elsewhere.

6.5 Subject term (key word) listing.

Bond
Primer

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – AR
Navy - AS
Air Force – 20

Preparing activity:
Army - AR
(Project 8040-2018-002)

Review activity:
Army - MD

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